Math 118: Mathematical Methods of Data Theory

Lecture 2: Solving Linear systems and Least Squares

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Linear Systems and Least Squares

- We want to solve the following equation for $\mathbf{x}$:

$$A\boldsymbol{x} = \boldsymbol{b} \tag{1}$$

  where $A \in \mathbb{R}^{m \times n}$ and $\boldsymbol{b} \in \mathbb{R}^m$.

- *Question: When does this system have at least one solution? When does it have a unique solution?*
- If $m = n$ and A is **nonsingular** then (1) has a unique solution.
- If $m < n$ then (1) is **underdetermined**. Discuss this later (maybe).
- If $m > n$ then (1) is **overdetermined** and has no solutions[1]. Solve **least squares problem** instead:

$$\min_{\boldsymbol{x} \in \mathbb{R}^m} \|A\boldsymbol{x} - \boldsymbol{b}\|_2$$

[1] In general

Solving Linear Systems using LU decomposition

- Consider again $A\boldsymbol{x} = \boldsymbol{b}$ with $A \in \mathbb{R}^{n\times n}$ nonsingular.
- Suppose $A = LU$ with L lower triangular with 1's on diagonal and U upper triangular. Solving system would be "easy".

Example

Suppose $L = \begin{bmatrix} 1 & 0 & 0 \\ \ell_{21} & 1 & 0 \\ \ell_{31} & \ell_{32} & 1 \end{bmatrix}$ and $U = \begin{bmatrix} u_{11} & u_{12} & u_{13} \\ 0 & u_{22} & u_{23} \\ 0 & 0 & u_{33} \end{bmatrix}$

Then $A\boldsymbol{x} = \boldsymbol{b} \Leftrightarrow LU\boldsymbol{x} = \boldsymbol{b}$. Let $\boldsymbol{y} = U\boldsymbol{x}$ then $L\boldsymbol{y} = \boldsymbol{b}$.

Strategy: Solve for $\boldsymbol{y}$ in using forward substitution with L then solve for $\boldsymbol{x}$ using backward substitution with U.

$$\begin{bmatrix} 1 & 0 & 0 \\ \ell_{21} & 1 & 0 \\ \ell_{31} & \ell_{32} & 1 \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix} \Rightarrow \begin{cases} y_1 = b_1 \text{ and } y_2 = b_2 - \ell_{21}y_1 \\ \text{and } y_3 = b_3 - \ell_{31}y_1 - \ell_{32}y_2 \end{cases}$$

$$\begin{bmatrix} u_{11} & u_{12} & u_{13} \\ 0 & u_{22} & u_{23} \\ 0 & 0 & u_{33} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix} \Rightarrow \begin{cases} x_3 = \frac{y_3}{u_{33}} \text{ and } x_2 = \frac{1}{u_{22}}\left(y_2 - u_{23}x_3\right) \\ \text{and } x_1 = \frac{1}{u_{11}}\left(y_1 - u_{12}x_2 - u_{13}x_3\right) \end{cases}$$

Solving Linear Systems using LU decomposition

Theorem

Any A can be written as $A = P^{-1}LU$ where:

- *P is a* **Permutation matrix**.
- *L is lower triangular with 1's on diagonal.*
- *U is upper triangular.*

Definition

P is a **Permutation matrix** if:

1. All entries of P either 1 or 0.
2. Only one non-zero entry in every row of P.
3. Only one non-zero entry in every column of P.

Example

Which of these is a permutation matrix?

$$P_1 = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 1 & 1 \end{bmatrix} \quad P_2 = \begin{bmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix} \quad P_3 = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 1/2 & 0 \end{bmatrix}$$

Solving Linear Systems using LU decomposition

- PA permutes the rows of A:

  If $P = \begin{bmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}$ and $A = \begin{bmatrix} 2 & 3 & 5 \\ -11 & 4 & 1 \\ 0 & -5 & 3 \end{bmatrix}$ then: $PA = \begin{bmatrix} 0 & -5 & 3 \\ 2 & 3 & 5 \\ -11 & 4 & 1 \end{bmatrix}$

- Can find P, L and U using Gaussian elimination (with partial pivoting) *See next slide.*
- Total number of **operations required**: $O(n^3)$.
- **Uses:** Solving linear systems, finding A^{-1}, computing determinant.
- **Advantage:** Does not change right hand side of $A\boldsymbol{x} = \boldsymbol{b}$. Suppose need to solve:

  $$A\boldsymbol{x}_1 = \boldsymbol{b}_1 \quad A\boldsymbol{x}_2 = \boldsymbol{b}_2 \ldots A\boldsymbol{x}_m = \boldsymbol{b}_m$$

  Only need to do Gaussian elimination once to determine $A = P^{-1}LU$.

Example: Finding the LU decomposition

$$\text{Let } A = \begin{bmatrix} 1 & 2 & 5 \\ 3 & 6 & 2 \\ 5 & 0 & 1 \end{bmatrix} \xrightarrow{\text{pivot:}} PA = \begin{bmatrix} 5 & 0 & 1 \\ 3 & 6 & 2 \\ 1 & 2 & 5 \end{bmatrix} \text{ where } P = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix}$$

$$PA \xrightarrow[R_3 - 1/5 R_1]{R_2 - 3/5 R_1} A^{(1)} := \begin{bmatrix} 5 & 0 & 1 \\ 0 & 6 & 7/5 \\ 0 & 2 & 24/5 \end{bmatrix} = L_1^{-1} PA \text{ where } L_1^{-1} = \begin{bmatrix} 1 & 0 & 0 \\ -3/5 & 1 & 0 \\ -1/5 & 0 & 1 \end{bmatrix}$$

$$A^{(1)} \xrightarrow{R_3 - 1/3 R_2} U := \begin{bmatrix} 5 & 0 & 1 \\ 0 & 6 & 7/5 \\ 0 & 0 & 65/15 \end{bmatrix} \text{ where } L_2^{-1} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & -1/3 & 1 \end{bmatrix}$$

Unwind:

$$U = L_2^{-1} A^{(1)} = L_2^{-1} L_1^{-1} PA \Rightarrow \underbrace{L_1 L_2}_{=L} U = PA$$

- Check that L is lower triangular.
- What about multiple pivots?

Least Squares: Motivation

Example

Suppose $A = \begin{bmatrix} 1 & 2 \\ -1 & 4 \\ 3 & 0 \end{bmatrix}$ and $\boldsymbol{b} = \begin{bmatrix} 3 \\ 9 \\ 5 \end{bmatrix}$ and consider $A\boldsymbol{x} = \boldsymbol{b}$ where $\boldsymbol{x} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$

$x_1 = -1$ and $x_2 = 2$ satisfies $1x_1+2x_3 = 3$ and $-x_1+4x_2 = 9$ but $3x_1+0x_2 \neq 5$

Conclusion: $A\boldsymbol{x} = \boldsymbol{b}$ has **no solutions**.

Idea: Minimize **residual error**:

- For any $\boldsymbol{x}$ define $\boldsymbol{r} = \boldsymbol{b} - A\boldsymbol{x}$.
- Let $\boldsymbol{x}^* = \underset{\boldsymbol{x}\in\mathbb{R}^m}{\arg\min} \|\boldsymbol{b} - A\boldsymbol{x}\|_2$
- Find $\boldsymbol{x}^*$.

If A is nonsingular then $\boldsymbol{x}^* = A^{-1}\boldsymbol{b}$.

Least Squares: The Normal Equations

- Let $f(\boldsymbol{x}) = \|\boldsymbol{b} - A\boldsymbol{x}\|_2^2$ then:

$$\boldsymbol{x}^* = \underset{\boldsymbol{x}\in\mathbb{R}^m}{\arg\min} \|\boldsymbol{b} - A\boldsymbol{x}\|_2 \Leftrightarrow \boldsymbol{x}^* = \underset{\boldsymbol{x}\in\mathbb{R}^m}{\arg\min} \underbrace{\|\boldsymbol{b} - A\boldsymbol{x}\|_2^2}_{=f(\boldsymbol{x})}$$

- *Homework exercise:* $\nabla f(\boldsymbol{x}) = A^\top(\boldsymbol{b} - A\boldsymbol{x})$.
- From Calc. 3: Find $\boldsymbol{x}^*$ by solving:

$$\left.\begin{aligned} 0 = \nabla f(\boldsymbol{x}^*) &= A^\top(\boldsymbol{b} - A\boldsymbol{x}^*) \\ \Rightarrow \quad A^\top A\boldsymbol{x}^* &= A^\top\boldsymbol{b} \\ \Rightarrow \quad \boldsymbol{x}^* &= \left(A^\top A\right)^{-1} A^\top\boldsymbol{b} \end{aligned}\right\} \text{Normal equation}$$

- Solving Normal equation can be **expensive** and **unstable**. Other methods for (approximately) finding $\boldsymbol{x}^*$ can be preferable.

Perturbation Analysis for Linear Systems

Consider again $A\boldsymbol{x} = \boldsymbol{b}$

Example

Suppose we construct A and $\boldsymbol{b}$ from real data. Noise/imprecision in measuring process means instead of A and $\boldsymbol{b}$ we obtain $A + \delta A$ and $\boldsymbol{b} + \delta \boldsymbol{b}$.

True solution satisfies: $A\boldsymbol{x} = \boldsymbol{b}$ where A and $\boldsymbol{b}$ are *unknown*

Our solution satisfies: $(A + \delta A)\hat{\boldsymbol{x}} = \boldsymbol{b} + \delta \boldsymbol{b}$ where $A + \delta A$ and $\boldsymbol{b} + \delta \boldsymbol{b}$ are *known*

Question: What is size of $\boldsymbol{x} - \hat{\boldsymbol{x}}$?

Theorem

1. *Define* **condition number** $\kappa(A) = \|A\|_2 \|A^{-1}\|_2$.
2. *Let* $r = \|\delta A\|_2 \|A^{-1}\|_2 < 1$.

Then: $$\frac{\|\boldsymbol{x} - \hat{\boldsymbol{x}}\|_2}{\|\boldsymbol{x}\|_2} \leq \frac{\kappa(A)}{1 - r} \left(\frac{\|\delta A\|_2}{\|A\|_2} + \frac{\|\delta \boldsymbol{b}\|_2}{\|\boldsymbol{b}\|_2} \right)$$

- Large condition number $\Rightarrow$ large amplification of noise.

Summary

1. A linear system $A\boldsymbol{x} = \boldsymbol{b}$ can be **underdetermined, overdetermined** or have a **unique solution**.
2. The LU **decomposition** is a useful way to keep track of Gaussian elimination for multiple uses.
3. **Least squares** is a way to find the "best approximate solution" to an overdetermined linear system.
4. The **condition number** of A dictates how sensitive $A\boldsymbol{x} = \boldsymbol{b}$ is to perturbation/error.